

Basic Electric Circuits

ENGS152 Circuits Lab

14 February 2026

Step 1: Simple LED Circuit

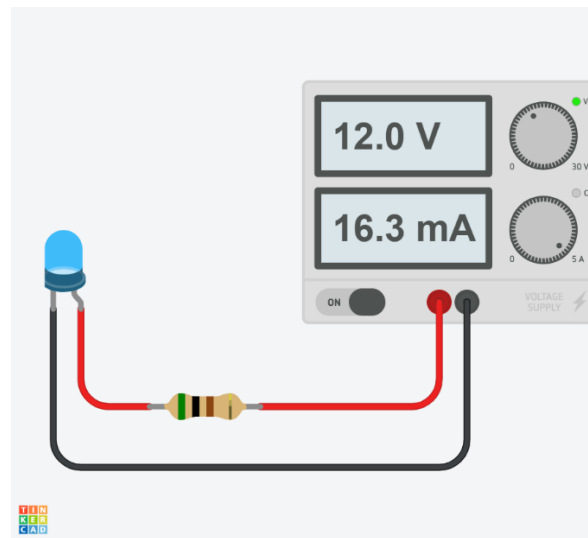
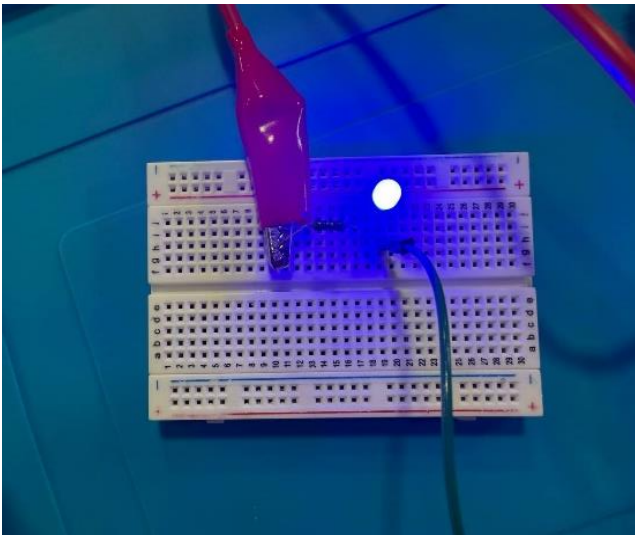
In this step we are required to build a simple circuit that will light up the LED. It is required to keep the voltage supply of the circuit to 12 V. However, no LED can handle such high voltages without connecting a resistor in series. Out of curiosity blue LED has been selected for the circuit. It's corresponding voltage drop of 2.5 – 3.7 V has been considered in the calculations to make sure not to damage the LED. It is also known that a LED can withstand 10 mA – 30 mA of current. Average values of were selected for the calculation.

After applying Kirchhoff's law, we derive that:

$$V_{resistor} = V_{supply} - V_{LED} = 12\text{ V} - 3.1\text{ V} = 8.9\text{ V}$$

$$R = \frac{V_{resistor}}{I} = \frac{8.9}{0.02} = 445\ \Omega$$

It is visible that $R = 445\ \Omega$ is the safe resistor value to not damage the LED, however the next closest resistor value available in the lab was 505 Ω , so the next resistor value has been used.



Step 2: Current measurement

To estimate the current in the circuit, we must use a multimeter. Multimeter in the circuit is connected in series, and the current measuring mode is selected. The connection diagram and a picture from the lab are provided below. It is visible that the current in the circuit was 17.90 mA, which does not exceed the amount of current a LED can withstand.



Step 3: LED Brightness Control

To change the brightness of the LED in the circuit we must regulate the amount of current flowing through the circuit. This is where a potentiometer comes to help. It is a simple electrical component that allows us to regulate its resistance. Increasing the resistance reduces the current and dims the LED, while decreasing the resistance increases the current and brightness the LED.

$$R_{min} = \frac{V}{I_{max}} = \frac{8.9}{0.02} = 445 \, \Omega, \quad R_{max} = \frac{V}{I_{min}} \approx \frac{8.9}{0.001} \approx 8.9 \, \text{k}\Omega.$$

Notice that the resistance must not go below $445\ \Omega$ to keep the current at safe levels. However, there is not upper bound on the potentiometer resistance value, because as large as we go, the larger the resistance, and the dimmer the light. $1\ \text{k}\Omega$ potentiometer has been used for this experiment, since it is the next convenient value potentiometer available in the lab after the value we calculated in theory. 3 pins of the potentiometer were used so it acts as a voltage divider compared to the 2-pin option which is a variable resistor. Below you can see the connection diagram and picture of LED at different potentiometer settings.

